# Open Design Project: Removal of Spotted Lanternflies with Compressed Air Canister

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## Introduction

As grape crops are mechanically harvested, spotted lanternflies (SLFs) that cling to the crops also enter the harvester. One or two SLFs can contaminate a batch, making the product of those grapes unsellable, wasting resources and increasing cost. There is currently no reliable mechanism to stop SLFs from entering the harvester.

Our goal is to design a harvester-mounted mechanism that removes SLFs before they enter the harvester. This would minimize interference with the harvesting process, reduce labor costs, and tackling the problem at this stage would give us the most control over the removal of SLFs in grape harvesting.

## Proposed Solution

We designed a compressed air mechanism that releases a concentrated stream of air to blow SLFs off grape vines. Our final design consists of 2 main subsystems:

*Trigger Actuation System*
- A crank handle rotates a 3D printed shaft that presses the compressed air canister trigger to release air to blow off the SLFs.

*Gimbal System*
- A 2 axis gimbal allows the cannister to remain stable while attached to the harvester and maintain aim at the SLFs on the grapevines

*Key features of our design include:*
- 3D printed shaft compresses the air-can trigger
- Crank handle provides actuation
- Acrylic housing holds the compressed air canister
- 2-axis Gimbal
- Compact design suitable for future harvesting integration

**First Prototypes:**

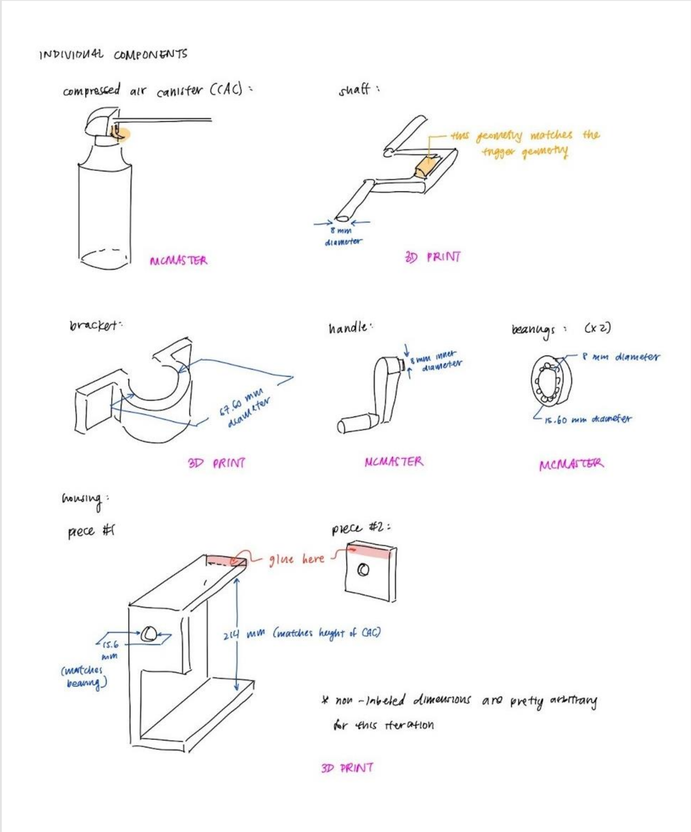

**Fig. 1**: Sketch of Design Components

**Fig. 2**: Functional Prototype

**Final Design:**

**Fig. 3**: CAD Render of Final Assembly

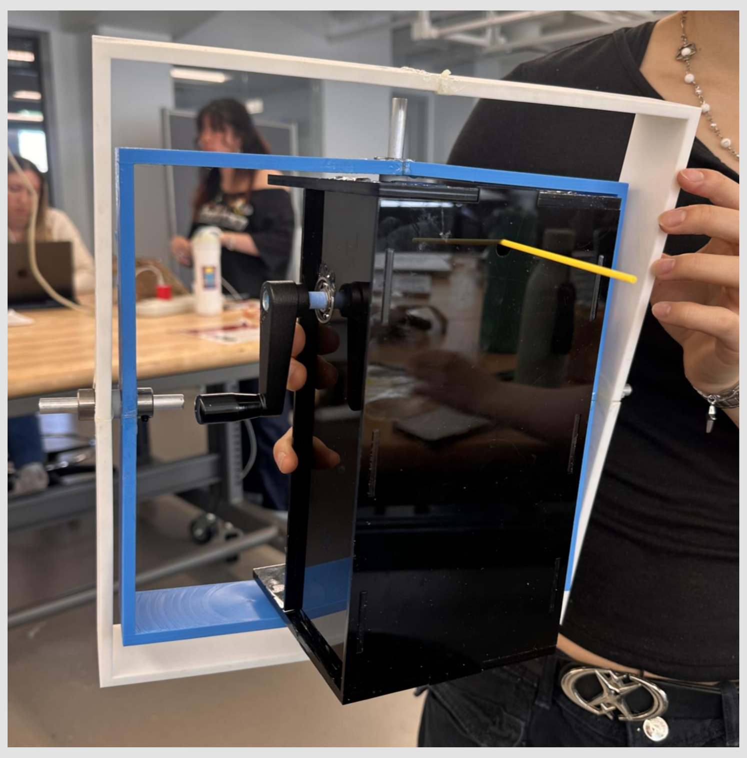

**Fig. 4**: Final Assembly with Cover

**Design Iteration**

Our first prototype tested the trigger compression mechanism. During testing, we found that the mechanism could actuate the trigger, but the 3D-printed shaft deflected under load, requiring more handle rotation before air was released. For our final prototype, we strengthened our shaft, replaced the temporary wood housing with acrylic and added a two-axis gimbal so the canister could aim at different target locations. This iteration shifted our design to a more complete removal system with both actuation and directional control.

## Testing

**Removal at Varied Distances: Blowing Modelled SLFs with Masking Tape from Various Distances**
- We created modelled SLFs with masking tape, weighing 1-5g each and used the compressed air canister to blow them around
- SLFs weigh an average of 0.3g, but have grip strength so we included a factor of 3
- Prototype successfully blows off 3D printed "SLFs" weighing around 1 gram at distances of up to 2.5 ft.

**Force Exerted by Shaft: Force Required to Compress the Trigger**
- Utilized a spring scale to measure the fore that the 3D printed shaft must exert on the trigger.
- To release a small stream of air, ~25N of force is needed. To fully depress the trigger, ~45N of force is necessary
- Our testing showed that our final shaft and crank mechanism was strong enough to consistently apply at least 45N without any significant deflection.

**Aiming Test: Gimbal and Canister Rotation**
- Our 2-axis gimbal stabilizes the cannister as it's mounted to the harvester, improving directional control and making the prototype compatible with the harvester process.

**Fig. 5**: Blowing Modelled SLFs from various distances

**Fig. 6**: Force test using spring scale

## Overall Results

Overall, through testing and design iterations, our prototype is successful in removing modelled SLFs in addition to two rotational degrees of freedom that stabilizes the cannister and allows for stable targeting of SLFs. Additionally, the design is compact and can easily be integrated into a harvester without disrupting the harvesting process or integrated into other larger mechanisms. We believe our design is promising and can be utilized along with other solutions to reduce the SLFs in the fields.

## Future Improvements

For future iterations of this prototype, several challenges remain to be addressed. One key challenge of the current design is the gimbal and the process of compressing the trigger on the air canister are separate mechanisms, making it difficult for both processes to proceed at the same time. Another challenge is that the compressed air canisters used are not refillable and will require frequent replacement. Additionally, a key risk that remains to be addressed is the potential for the compressed air canister to blow off grapes as well as SLFs in the process.

With these key risks and challenges in mind, here are some recommendations that we suggest for future iterations of this design:
- Implement electronics and motors that automate the movement of the shaft as well as the rotation of the gimbal to allow both mechanisms to be controlled simultaneously and reduce manual labor.
- Automate the SLF detection with sensors or cameras to avoid blowing off grapes, reduce energy consumption of the compressed air by actuating the mechanism only when SLFs are detected and increase operational efficiency.
- Integrate a compressed air gun that can be conveniently refilled and maintained.

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